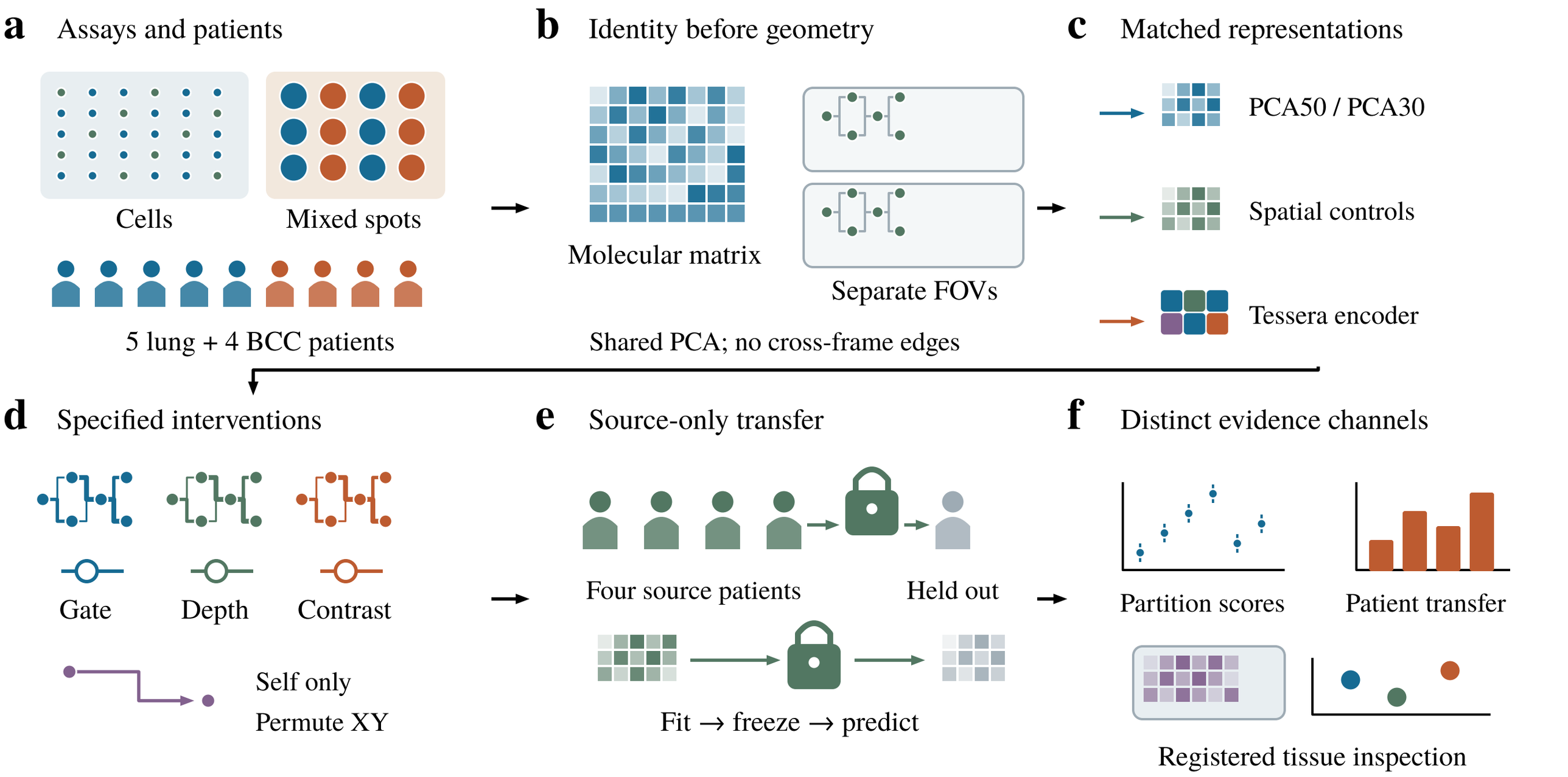

a
Assays and patients
Cells
Mixed spots
5 lung + 4 BCC patients
b
Identity before geometry
Molecular matrix
Separate FOVs
c
Matched representations
PCA50 / PCA30
Spatial controls
Tessera encoder
Shared PCA; no cross-frame edges
d
Specified interventions
Gate
Depth
Contrast
Self only
Permute XY
e
Source-only transfer
Four source patients
Held out
Fit → freeze → predict
f
Distinct evidence channels
Partition scores
Patient transfer
Registered tissue inspection